

RIPHAH INTERNATIONAL UNIVERSITY



Faculty of Computing

FINAL YEAR PROJECT PROPOSAL & PLAN

CareLink: ML-Based Medication Adherence and Chronic-Care Coordination System

Project Team

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CareLink

Change Record

Author(s)	Version	Date	Notes	Supervisor's Signature
Saleem,Saif,Siyab	1.0		Original Draft	
			Changes Based on Feedback From Supervisor	
			Changes Based on Feedback From Supervisor	
			Changes Based on Feedback From Supervisor	

Project Proposal

Project Title: CareLink: ML-Assisted Medication Adherence and Chronic-Care Coordination System

Opportunity & Stakeholders:

Medication adherence is an important part of long-term chronic disease management. Patients may miss scheduled doses, take medication late, or repeatedly report medication events in ways that are difficult to review manually. This creates an opportunity for a software system that goes beyond fixed reminders by recording medication events, analyzing adherence-related patterns, identifying increasing adherence risk, and supporting appropriate follow-up.

CareLink will focus on two chronic conditions: hypertension and diabetes. The system will combine a patient-facing smartphone application with a web-based care-coordination portal. The main stakeholders are patients, caregivers, care coordinators, authorized healthcare personnel, and system administrators. CareLink will support human decision-making and will not independently diagnose conditions, prescribe medicines, or change treatment plans.

Existing System/ Description of the Current Situation:

Existing medication-management applications provide functions such as medication schedules, reminders, medication-event logging, adherence statistics, notifications, caregiver access, and reports. These systems demonstrate that digital medication management is feasible, but reminderbased functionality alone does not necessarily identify changing adherence patterns or determine when additional follow-up may be required.

The current situation can broadly be considered at three levels. Basic systems focus on reminding patients when medication is due. More advanced systems record medication history and provide adherence statistics or caregiver visibility. Predictive approaches can use historical behavior to identify patterns associated with increased adherence risk.

CareLink will investigate the integration of these capabilities into a focused chronic-care workflow for hypertension and diabetes. The system will combine medication management, adherence-related analysis, a project-specific machine-learning risk model, predefined support

rules, care coordination, and reporting rather than treating medication reminders as the complete solution.

Problem Statement:

Patients managing chronic diseases may need to follow long-term medication schedules involving different medicines, dosages, frequencies, and timings. Over time, they may miss doses, take medication late, or develop repeated patterns of non-adherence. A conventional reminder can notify a patient that medication is due, but it does not necessarily identify when repeated adherence problems are developing or when follow-up may be useful.

Continuous forwarding of every missed-dose event to caregivers or care coordinators may create unnecessary notifications and increase manual workload. Therefore, CareLink addresses how medication-taking information can be systematically recorded, analyzed, and used to estimate adherence risk while providing appropriate support and human follow-up. Patient-reported medication events will be treated as reported information; the system will not claim to directly verify that a medicine was swallowed.

Proposed Solution:

CareLink is proposed as an ML-assisted medication adherence and chronic-care coordination system consisting of a smartphone application and a web-based portal. The patient application will manage profiles, selected chronic conditions, medications, schedules, reminders, medication-event recording, and adherence-related history. The web portal will allow authorized caregivers and care coordinators to monitor assigned patients, review adherence-related trends, receive relevant alerts, record follow-up actions, and generate reports.

The central technical component will be a project-specific machine-learning model developed and evaluated by the project team. Using suitable adherence-related features, the model will estimate an adherence-risk level such as Low, Moderate, or High. A separate rule-based layer will control reminders, alerts, and follow-up actions according to predefined conditions. The model will not diagnose disease, prescribe medication, modify treatment, or make independent clinical decisions. The final model, features, and dataset will be selected and documented according to feasibility and available project data

Scope of the Project:

The committed scope of CareLink is organized into the following core modules. Each module represents a functional area of the system and will be further decomposed into development activities within the Work Breakdown Structure. The project is intentionally limited to hypertension and diabetes so that the team can study the selected domain in sufficient depth.

1. Patient & Profile Management

This module manages patient profiles and essential information needed by CareLink. It allows authorized users to maintain patient details, selected chronic conditions, medication-related information, and care-related records. The module provides the information required by other CareLink components while applying appropriate access controls.

2. Medication Management

This module manages the medications recorded for each patient. It supports medication names, dosage information, frequency, schedules, and relevant instructions. The schedules are used by the reminder, medication-event recording, adherence analysis, and reporting components.

3. Adherence Tracking & Analysis

This module records patient-reported medication events, including reported taken, missed, delayed, and unconfirmed events. It calculates indicators such as reported adherence rate, misseddose frequency, delay patterns, and recent trends. These indicators are used to identify changes in reported medication-taking behavior. The system does not treat a patient confirmation as proof that the medicine was swallowed.

4. Health Data Monitoring

This module allows selected health information relevant to hypertension and diabetes to be recorded by the patient or an authorized user. Examples may include blood-pressure or bloodglucose readings when available to the project workflow. The module presents recorded information alongside medication and adherence-related data and does not provide autonomous diagnosis.

5. AI/ML Risk Prediction

This module contains the project machine-learning component for estimating medication adherence risk. The model will use relevant recorded features, such as missed or delayed medication events, reported adherence trends, medication frequency, and other suitable project features. The output will be a risk estimate to support human review, not a medical diagnosis or proof of medication ingestion.

6. Personalized Reminders & Intervention

This module provides medication reminders and predefined support actions based on medication schedules and recorded adherence behavior. Reminders can be delivered for upcoming or missed medication events, while predefined rules can trigger suitable follow-up when adherence-risk conditions are detected. The module does not change medication or treatment plans.

7. Caregiver / Care Coordinator Dashboard

This module provides a centralized web-based dashboard for caregivers and care coordinators to monitor assigned patients. It presents relevant medication schedules, reported adherence patterns, risk estimates, alerts, and patient-related information in an organized manner. Authorized users can review the available information and record appropriate follow-up actions.

8. Alerts & Notifications

This module manages notifications related to medication schedules, missed or delayed reported events, elevated adherence risk, and other defined care events. Notifications are generated according to predefined conditions and can be directed to patients, caregivers, or care coordinators according to their roles. The module is designed to avoid unnecessary repeated alerts.

9. Reports & Analytics

This module provides reports and analytical views of recorded medication events and care coordination activity. Reports may include reported adherence rates, missed and delayed events, adherence-related trends, risk-level history, unresolved alerts, and follow-up records. Reporting is intended to help authorized users review patient patterns without claiming clinical validation.

10. Secure Authentication & Data Privacy

This module provides secure user authentication, authorization, and access control for CareLink. It ensures that patients, caregivers, care coordinators, and administrators can access only the information and functions appropriate to their roles. The module also addresses protection of sensitive patient information through secure data handling, controlled access, and appropriate privacy measures.

Smartphone Application

The patient-facing mobile application will provide login/profile functionality, selected disease and medication information, medication schedules, reminders, medication-event recording, reported adherence history, notifications, and appropriate support information.

Web Portal

The web portal will provide authorized interfaces for caregivers, care coordinators, and administrators. Its main functions will include patient monitoring, adherence-related analytics, risk information, alerts, reports, follow-up history, and care-coordination functions.

Selected Disease Scope

The initial project scope will cover:

- **Hypertension**
- **Diabetes**

Medication and adherence requirements for hypertension and diabetes will be studied and used to guide the system requirements, data structure, workflows, and adherence-management features.

Out of Scope

The committed FYP implementation will not include autonomous diagnosis, prescription, treatment modification, hospital EMR integration, pharmacy-system integration, clinical deployment, mandatory specialized hardware, smart pill-box hardware, IoT-based medication verification, computer-vision-based pill-intake verification, continuous camera monitoring, or replacement of complete medical records.

Computer vision and specialized medication-monitoring hardware are outside the committed FYP scope. They may be considered only as future extensions after the core software and machinelearning components are completed and evaluated.

List of Faculty Proposed Changes

Project Title: CareLink

Supervisor's Signature: _____

Proposed Change	Proposed By	Supervisor's Decision
	Name of Faculty Member(s) who proposed this change	Approved/Disapproved and/or Comments
Good idea	muhammad.abdullah@riphah.edu.pk	
Make AI model	imran.fc@riphah.edu.pk	Approved
NO Agents are required in the project, computer vision may be used for monitoring of the critical patients and strong reporting must be introduced	sharjeel.gilani@riphah.edu.pk	
Select a set of chronic diseases / critical conditions and study their medication/treatment adherence requirements. Design the system around monitoring and supporting adherence management for the selected diseases.	zeeshan.fc@riphah.edu.pk	Approved

Suggestions: • The scope of the project (product features) is required to be welldefined Currently, it is too open. Comments: • Domain knowledge of the students is good. • Use of IOT devices for data input is recommended • The students committed to develop a smartphone And application Web portal	sajid.qureshi@riphah.edu.pk	
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Reasoning for Suggestions Approval/DisApproval

1. Make AI Model — Approved

Reasoning: The project team approved this suggestion and decided to develop and evaluate a project-specific machine-learning model for medication-adherence risk estimation. The model will use suitable recorded adherence-related features and will be evaluated using appropriate performance metrics. The final model, features, and dataset will be selected according to project feasibility and documented during development.

2. No Agents Required — Approved

Reasoning: The team approved the removal of the previously proposed multi-agent architecture. The system will instead use a project-specific machine-learning model for adherence-risk estimation, while predefined rules will control reminders, alerts, and follow-up actions. This keeps the architecture focused and feasible while maintaining human oversight for care-related decisions.

3. Computer Vision May Be Used for Monitoring Critical Patients — Disapproved for Current Scope / Future Work

Computer Vision for Medication Intake Verification: Not included in the core implementation scope. The team reviewed camera-based medication-intake verification and identified requirements involving controlled camera positioning, lighting, medication visibility, action recognition, ingestion verification, dataset availability, model training, accuracy validation, and patient privacy. Reliable intake verification would require additional computer-vision expertise and validation beyond the current project scope, so the feature has been excluded from the committed FYP implementation.

4. Strong Reporting Should Be Introduced — Approved

Reasoning: The team approved this suggestion and retained reporting as a core component. The system will provide patient-level and care-team-level reports covering recorded adherence trends, missed and delayed events, risk history, alerts, and follow-up activity. System-level analytics beyond the defined project workflow are not required for the committed implementation.

5. Select a Set of Chronic Diseases / Critical Conditions — Approved

Reasoning: The team approved this suggestion and narrowed the disease scope to two chronic conditions: hypertension and diabetes. Medication-adherence requirements for these conditions will be studied and used to guide the system requirements, data structure, workflows, and adherence-management features. This provides a defined domain for deeper study while keeping the project manageable.

6. Define and Extend the Project Scope — Approved

Reasoning: The team approved this suggestion by defining the project boundaries. CareLink includes patient and medication management, medication-event recording, adherence-related analysis, machine-learning risk estimation, predefined support actions, caregiver and carecoordinator support, alerts, reporting, security, and administration. Autonomous diagnosis, treatment modification, computer-vision verification, and mandatory specialized hardware are excluded.

7. IoT Devices for Data Input — Deferred to Future Work

Reasoning: The team considered the recommendation but decided not to make IoT or specialized hardware a dependency of the current FYP. The committed system will use medication events and selected health information recorded through the smartphone application and web-based carecoordination workflow. No external medical device is required for the committed implementation.

Any specialized hardware or smart medication-monitoring device is outside the current FYP scope and will not be required for the final demonstration.

8. Smartphone Application and Web Portal — Approved / Retained

Reasoning: The team approved and retained the development of both interfaces. The smartphone application will support patients with medication schedules, reminders, medication-event recording, and adherence-related information, while the web portal will support authorized caregivers, care coordinators, and administrators through monitoring, alerts, reports, and carecoordination functions.

Project Plan

1. Project Management

1.1 Work Breakdown Structure (WBS)

The WBS will serve as the primary planning and tracking structure for CareLink. It decomposes the project into functional modules and supporting activities so that each committed requirement can be assigned to a responsible team member. The WBS will cover the mobile application, web portal, backend services, database, machine-learning model, intervention workflow, reporting system, security, testing, and documentation.

The module-based structure also provides traceability between the proposal and implementation. A module included in the approved scope must have corresponding requirements, design elements, implementation activities, testing activities, and final documentation. This approach reduces the possibility of promising functionality in the proposal that is not represented in the actual development plan.

1.2 Roles & Responsibility Matrix

Ownership will be distributed across the three-member team according to each member's primary technical responsibilities. One member may primarily manage the mobile and patient-facing functionality, another may focus on backend, adherence analysis, and machine learning, while the third may focus on the web portal, reporting, security, and administration. Exact ownership will be finalized according to the team's actual member assignments.

Although each module will have a primary owner, major integration, testing, documentation, and final validation activities will remain shared responsibilities. This ensures that the project is not dependent on a single member for understanding the complete system.

1.3 Change Control System

Any proposed change to the approved project scope, architecture, technology stack, disease scope, or major functionality will be documented rather than adopted informally. Changes may originate from supervisor feedback, committee recommendations, requirements analysis, technical feasibility findings, or internal project decisions.

Each significant change will record the date, proposed modification, reason for the change, impact on existing modules, and approval status. Changes affecting the committed scope will only be incorporated after appropriate review by the project team and supervisor. This provides traceability between the original proposal, approved modifications, and final implemented system.

2. Reports / Documentation

2.1 Final Documentation Introduction

The final documentation will introduce CareLink, its motivation, selected chronic conditions, target stakeholders, and the medication-adherence problem addressed by the project. It will describe the transition from conventional reminder-based medication management toward structured adherence-related monitoring and machine-learning-supported risk estimation.

The introduction will also establish the system boundaries, particularly the distinction between adherence-risk estimation and clinical decision-making. It will explain that CareLink is a softwarebased support system rather than a replacement for healthcare professionals or clinical systems.

2.2 Literature / Market Survey

The literature survey will review research related to medication adherence, chronic disease management, digital medication-management systems, adherence prediction, machine-learning methods, and care coordination. The selected hypertension and diabetes domains will be studied to identify relevant medication-adherence requirements and domain considerations.

The market survey will examine existing medication-management applications and related digital health platforms to identify their major functions, limitations, and differences from the proposed CareLink workflow. Primary research studies and reliable sources will be prioritized for technical and medical claims, while existing applications will be evaluated based on documented functionality.

2.3 Requirements Analysis

Requirements will be developed from literature, domain guidance, stakeholder discussions, and analysis of existing medication-management workflows. Functional requirements will cover patient management, selected diseases, medications, schedules, medication events, risk estimation, predefined support actions, notifications, caregiver access, care coordination, reporting, and administration.

Non-functional requirements will address security, privacy, usability, performance, reliability, maintainability, and access control. The requirements document will also explicitly record scope exclusions such as autonomous diagnosis, prescription modification, specialized hardware, computer vision, and clinical deployment.

2.4 System Design

System design documentation will describe the CareLink architecture and the relationship between the mobile application, web portal, backend services, database, machine-learning component, intervention engine, and reporting system.

It will include architecture diagrams, database design, API structure, user-role relationships, system workflows, and relevant UML diagrams. The ML design will document feature preparation, model selection, training, evaluation, prediction output, and how predictions are passed to the predefined intervention layer.

Design decisions such as selecting Flutter for the mobile application, Angular for the web portal, FastAPI for backend services, PostgreSQL for relational storage, and Python/scikit-learn for machine learning will be documented with appropriate justification.

2.5 Implementation

The implementation section will document the development of the CareLink modules according to the dependency-based WBS. It will cover the patient mobile application, web portal, backend APIs, database, medication and adherence-related workflow, ML model, intervention and notification logic, reporting, authentication, and administration.

The ML implementation will document dataset preparation, preprocessing, feature engineering, model training, model comparison, evaluation, and integration into the application. The final

dataset and model will be selected according to availability and feasibility, and implementation limitations will be recorded honestly.

2.6 Testing & Performance Evaluation

Testing will be conducted at multiple levels, including unit testing, API testing, integration testing, frontend testing, database testing, security-related testing, and end-to-end system testing. The machine-learning component will be evaluated using appropriate metrics such as precision, recall, F1-score, and ROC-AUC where applicable. Model evaluation will focus on the defined project prediction task and available dataset rather than claiming clinical validation or direct verification of medication ingestion.

System evaluation will also consider response time, notification behavior, report generation, rolebased access, data consistency, and overall workflow correctness. Testing results will provide the evidence used to assess whether the implemented system satisfies its defined software requirements.

2.7 Conclusion & Outlook

The conclusion will evaluate CareLink against its approved objectives and summarize the implemented medication-management, adherence-related analysis, machine-learning, intervention, coordination, and reporting capabilities.

Limitations will be reported honestly, including dataset limitations, patient-reported medication events, lack of direct ingestion verification, lack of clinical deployment, absence of hospital or pharmacy integration, and exclusion of specialized medication hardware and computer vision from the committed implementation.

Future work may include integration with validated healthcare systems, larger real-world datasets, additional chronic conditions, specialized medication-monitoring hardware, computer-visionbased verification, and further model improvement subject to appropriate clinical, privacy, and technical validation.

2.8 End User Documentation

End-user documentation will be prepared according to the main system roles. Patient documentation will explain account access, medication management, schedules, reminders, medication-event recording, and viewing adherence-related information.

Caregiver documentation will explain authorized patient access, adherence monitoring, alerts, and relevant reports. Care coordinator documentation will cover patient assignment, risk information, follow-up records, and report generation. The documentation will use simple language and avoid unnecessary technical terminology.

2.9 Application Administration Documentation

Application administration documentation will explain administrative functions available through the web portal. It will cover user and role management, authorization management, patientcaregiver relationships where applicable, system configurations, access review, and administrative monitoring.

It will also describe how administrators can review system activity and maintain appropriate access to protected information without directly modifying clinical treatment decisions.

2.10 System Administrator Documentation

System administrator documentation will cover deployment and maintenance of the FastAPI backend, PostgreSQL database, Angular web application, and supporting services. It will document environment configuration, database setup, application configuration, authenticationrelated settings, logging, backups, recovery procedures, and routine maintenance.

The documentation will also explain deployment of the machine-learning component and the process required to update a trained model after appropriate evaluation.

3. System

3.1 Development Environment

3.1.1 IDE

Visual Studio Code will be used as the primary development environment for backend, web, machine-learning, database-related, and general project development. Android Studio may be used where required for Android-specific development or testing.

The selected development tools provide support for the project's programming languages, frameworks, debugging, extensions, package management, and Git-based workflows.

3.1.2 Version Control

Git and GitHub will be used for source-code version control and collaborative development. Team members will work using feature-based branches where appropriate, followed by code review and controlled merging into the shared project branch.

Version control will also provide a history of significant implementation changes and allow the team to recover previous versions when required.

3.1.3 Server

The backend will be developed using Python with FastAPI. FastAPI will provide RESTful APIs connecting the mobile application and web portal with the database, authentication system, medication-management modules, adherence-analysis components, intervention logic, reporting functions, and ML prediction service.

authentication system, medication-management modules, adherence-analysis components, intervention logic, reporting functions, and ML prediction service.

The project may use a local development server during implementation, with deployment configuration finalized according to project requirements.

3.1.4 Database

PostgreSQL will be used as the primary relational database. It will store structured information including users, roles, patients, diseases, medications, schedules, medication events, adherence indicators, risk results, alerts, interventions, and relevant audit information.

A relational database is appropriate because the system contains interconnected entities such as patients, medications, schedules, adherence events, care relationships, alerts, and intervention records.

3.2 Presentation Layer

The following are the Presentation layer:

3.2.1 Deliverable 1 — Patient Smartphone Application

The patient-facing application will be developed using Flutter. It will provide a mobile interface for patient registration/login, profile management, selected disease and medication information, medication schedules, reminders, medication-event recording, reported adherence history, notifications, and appropriate support information.

The application will prioritize simple navigation and readable interfaces because patients may interact with it repeatedly during their daily medication routine.

3.2.2 Deliverable 2 — Caregiver Portal

The caregiver interface will provide authorized users with a focused view of patients who have granted appropriate access. It will display reported adherence information, missed-dose patterns, relevant alerts, and adherence trends without exposing unnecessary patient information. patterns, relevant alerts, and adherence trends without exposing unnecessary patient information.

The interface will allow caregivers to understand the patient's adherence status without requiring them to manually inspect every individual medication event.

3.2.3 Deliverable 3 — Care Coordinator Dashboard

The care coordinator dashboard will provide a broader monitoring interface for assigned patients. It will include patient lists, adherence trends, risk indicators, active alerts, follow-up history, and report access.

The dashboard will emphasize prioritization and summarized information so that care coordinators can identify patients requiring attention.

3.2.4 Deliverable 4 — Reporting & Visualization Interface

The presentation layer will include visual reporting components for reported adherence trends, risk history, missed-dose patterns, alert activity, and follow-up records.

Charts, tables, summaries, and filters will be used where appropriate to make the collected information easier to interpret.

3.3 Business Logic Layer

The business logic layer is as follows:

3.3.1 Deliverable 1 — Authentication & Authorization

This component will manage secure login, user identity, role-based access, and authorization rules. Different system roles will receive access only to functions and information appropriate to their permissions.

3.3.2 Deliverable 2 — Patient, Disease & Medication Management

This component will implement the business rules for patient profiles, selected diseases, medications, dosages, frequencies, schedules, and medication status.

It will ensure that medication information is stored consistently and can be used by the reminder and adherence modules.

3.3.3 Deliverable 3 — Medication Scheduling & Reminder Engine

The reminder engine will process medication schedules and generate appropriate reminders for patients. Medication-event outcomes will be linked with the corresponding scheduled medication event so that reported adherence indicators can be calculated.

3.3.4 Deliverable 4 — Adherence Tracking & Analysis

This component will process medication events and calculate adherence-related indicators such as reported adherence rate, missed doses, delayed doses, missed-dose streaks, and recent trends. These indicators will provide the primary input for the machine-learning risk-estimation component.

These indicators will provide the primary input for the machine-learning prediction component.

3.3.5 Deliverable 5 — Machine-Learning Risk Prediction

The ML component will be implemented using **Python and scikit-learn** or another justified machine-learning library selected during development.

The module will perform data preprocessing, feature preparation, model training, evaluation, and prediction. Candidate models may include Logistic Regression, Random Forest, and other suitable supervised-learning algorithms. The final model will be selected based on the project dataset and evaluation results.

The output will represent adherence-related risk rather than a medical diagnosis or proof of medication ingestion.

3.3.6 Deliverable 6 — Intervention & Notification Management

This component will apply predefined rules to adherence indicators and ML risk outputs. It will determine whether the appropriate response is a normal reminder, additional support, caregiver notification, or care-coordinator follow-up. The intervention engine will not modify medication or treatment plans.

The intervention engine will operate under predefined safety constraints and will not modify medication or treatment plans.

3.3.7 Deliverable 7 — Care Coordination

This component will manage caregiver and care-coordinator relationships, assigned patients, follow-up actions, intervention history, and relevant coordination activities.

It will provide the connection between automated system alerts and human follow-up.

3.3.8 Deliverable 8 — Alerts & Notifications

The notification component will generate and manage alerts resulting from configured conditions. It will distinguish routine medication reminders from higher-priority adherence or risk-related notifications.

Notification history will be retained to support reporting and evaluation of intervention workflows.

3.3.9 Deliverable 9 — Reporting & Analytics

The reporting engine will generate patient-level and care-team-level reports from stored medication events, adherence indicators, risk, alert, and follow-up data. It will support summaries such as adherence-related trends, risk history, patients requiring attention, active alerts, and followup activity.

It will support summaries such as adherence trends, risk history, patients requiring attention, active alerts, and intervention activity.

12.4 Data Management Layer

3.34.1 Deliverable 1 — User & Role Data

The database will maintain authenticated user information, system roles, permissions, and relevant account status.

3.4.2 Deliverable 2 — Patient & Disease Data

Patient profiles and selected chronic-condition information will be stored in structured relational tables with appropriate relationships and access controls.

3.4.3 Deliverable 3 — Medication & Schedule Data

Medication names, dosage information, frequency, schedules, duration, status, and related instructions will be stored for use by the medication-management and reminder modules.

3.4.4 Deliverable 4 — Medication Event & Adherence Data

The system will store medication events such as reported taken, missed, delayed, and unconfirmed events. These records will support adherence-related calculations, trend analysis, and ML feature generation.

3.4.5 Deliverable 5 — ML Data & Prediction Records

The system will maintain appropriate data required for model training and evaluation and store prediction outputs with relevant prediction timestamps. The selected dataset and feature set will be documented as part of the ML evaluation.

Where explanations or supporting factors are implemented, they will be retained with the prediction record where appropriate.

3.4.6 Deliverable 6 — Alert & Intervention Data

The database will store generated alerts, notification status, intervention actions, follow-up records, and relevant outcomes. This will allow the system to produce intervention-history reports and evaluate the notification workflow.

Deliverable 7 — Audit & Activity Data

Relevant security and system activities will be logged to support accountability, troubleshooting, and review of access to sensitive information.

3.5 Physical Layer

3.5.1 Deliverable 1 — Patient Mobile Devices

Standard Android smartphones will be used to develop, test, and demonstrate the patient-facing Flutter application. No specialized medical hardware will be required for the committed implementation.

3.5.2 Deliverable 2 — Web Client Devices

Standard computers or laptops will be used to access the Angular-based caregiver, care coordinator, and administrator web portal through a supported browser.

3.5.3 Deliverable 3 — Application Server

The FastAPI backend will operate on a development or deployment server responsible for handling API requests, authentication, business logic, adherence processing, intervention workflows, and communication with the database and ML components.

3.5.4 Deliverable 4 — Database Server

The PostgreSQL database will provide persistent storage for CareLink's structured application data. Database deployment will be configured according to the final project environment and demonstration requirements.

3.5.5 Deliverable 5 — ML Execution Environment

The machine-learning component will run in the project's Python environment and will provide predictions to the backend through the defined application architecture.

Project Scope Boundary

CareLink will focus on a defined chronic-care domain rather than attempting to solve medication adherence for all diseases. The committed implementation will cover hypertension and diabetes through patient and medication management, medication scheduling, medication-event recording, adherence-related analysis, project-specific machine-learning risk estimation, predefined support actions, caregiver and care-coordinator coordination, alerts, reporting, and security controls. Specialized hardware, IoT medication verification, computer-vision-based intake verification, autonomous diagnosis, and treatment modification are excluded from the current FYP implementation.

Roles & Responsibility Matrix:

The purpose of roles & responsibility matrix is to identify who will do what.

WBS #	WBS Deliverable	Activity #	Activity to Complete the Deliverable	Duration (Days)	Responsible Team Member(s) & Role(s)
1.2	Requirements & Research	1.2.1	Literature review and existing adherence systems	5	Saif Majid Khan (Research & Backend)
1.2	Requirements & Research	1.2.2	Market survey and domain consultation	4	Siyab Ahmad Khan (Research & Web)
1.2	Requirements & Research	1.2.3	Functional and non-functional requirements	6	All Members (Requirements)
1.3	System Design	1.3.1	System architecture and module interaction	4	Saif Majid Khan (System Architecture)
1.3	System Design	1.3.2	Database schema and data-flow design	4	Saif Majid Khan (Database)
1.3	System Design	1.3.3	Mobile and web UI/UX design	4	Saleem Aman & Siyab Ahmad Khan (UI/UX)

1.3	System Design	1.3.4	ML workflow and risk prediction design	3	All Members (ML/QA)
1.4	Mobile Application	1.4.1	Patient authentication and profile	4	Saleem Aman (Mobile)
1.4	Mobile Application	1.4.2	Medication schedules and reminders	6	Saleem Aman (Mobile)
1.4	Mobile Application	1.4.3	Medication-event recording and adherence tracking	6	Saleem Aman (Mobile)

Approval

Project Supervisor

Comments _____

Name: _____

Date: _____ Signature: _____

Project Coordinator

Comments _____

Name: _____

Date: _____ Signature: _____